University of Science
Faculty of Information and Technology

Physics for Information Technology
Course Project

Proposal: Smart Shrimp Pond

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1 | Introduction

1.1 | Acknowledgement

We would like to express our sincere gratitude to our lecturer and instructors of the *Physics for Information Technology* course for their valuable guidance, insightful feedback, and continuous support throughout the duration of this project. Their academic instruction and mentorship were instrumental in helping us translate theoretical physics and electronics concepts into a practical Internet of Things (IoT) engineering solution.

Additionally, we wish to thank our classmates and peers in Class **24C06** for their constructive discussions and cooperation during the prototype development and testing phases.

1.2 | Group Information

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1.3 | Project Information

- **Course:** Physics for Information Technology
- **Class:** 24C06
- **Project Name:** Smart Shrimp Pond – Integrated IoT Sensors & Emergency Response System
- **Project Category:** Warning System, Emergency Response, IoT Sensors

2 | Background and Motivation

2.1 | Background: Aquaculture scale and shrimp mortality

Brackish water shrimp farming—primarily Whiteleg (*Litopenaeus vannamei*) and Black Tiger shrimp (*Penaeus monodon*)—is a major economic pillar in Southeast Asia. In Vietnam's Mekong Delta, provinces like Ca Mau account for nearly 700,000 ha of farming area [1]. As farms transition toward intensive and super-intensive models (stocking densities > 100–300 individuals/m^2), operational complexity and susceptibility to environmental shocks increase dramatically [2].

Despite high yield potentials, the industry suffers annual losses of **25% to 35%** from water quality degradation and infectious outbreaks. The most vulnerable period occurs during the **first 4 to 6 weeks (Days 1–45 post-stocking)** [2]:

- **Environmental Vulnerability:** Immature post-larvae (PL) and juveniles suffer severe immunosuppression when water chemistry (pH, Dissolved Oxygen [DO], salinity, temperature) fluctuates rapidly [3].
- **Early Mortality Syndrome (EMS/AHPND):** Striking within 30 days post-stocking, pathogenic *Vibrio parahaemolyticus* proliferation is heavily accelerated by hypoxia (DO $<$ 4.0 mg/L) and sediment accumulation [3], [4].
- **Manual Monitoring Limitations:** Conventional manual chemical kits (sampled 1–2 times daily) fail to capture nocturnal hypoxia or sudden rain-induced acidification, leaving ponds exposed to catastrophic die-offs.

2.2 | Motivation: Empowering farmers through automated workflows

The **Smart Shrimp Pond** system alleviates the heavy physical and psychological burden on farmers by replacing manual water testing, paddlewheel operation, and manual feeding with an IoT monitoring and control architecture. The system executes **three closed-loop workflows** to address primary operational hazards:

1. **Rainfall & Acidification Mitigation (Stratification Control):**
   - *Challenge:* Heavy rain introduces acidic freshwater (pH $<$ 6.0), forming a surface layer that causes salinity stratification and blocks atmospheric oxygen exchange.
   - *Response:* Rain and ultrasonic water-level sensors trigger **surface discharge pumps** to drain excess surface water and **paddlewheels** to break stratification, while alerting the farmer via Email to prepare $CaCO_3$ liming.
2. **Nocturnal Hypoxia Prevention (DO Regulation):**
   - *Challenge:* Between 02:00 and 05:00 AM, phytoplankton respiration depletes DO below the 4.0 mg/L biological survival threshold [3].
   - *Response:* A continuous DO sensor monitors levels continuously; if DO $<$ 4.0 mg/L, the controller **deactivates automated feeders** and activates surface aeration (paddlewheels) at maximum capacity to rapidly restore oxygen levels until DO stabilizes above 5.5 mg/L.
3. **Extreme Heat & Salinity Control:**
   - *Challenge:* Midday heat (> 33°C) accelerates evaporation, spiking salinity (> 30 ppt) and inducing osmotic stress.
   - *Response:* Correlation of high temperature and rising conductivity triggers an **intake pump** to draw dilution water from a settling pond, logs the event to the cloud, and reduce or suspend automatic feeding.

3 | Product Scope

3.1 | Overall Data Flow

The controller reads all sensors and publishes a JSON telemetry message to MQTT. Node-RED subscribes to telemetry, displays data, evaluates higher-level workflows, stores data in Firebase, and sends control commands. The ESP8266 reports the actual output state after every command.

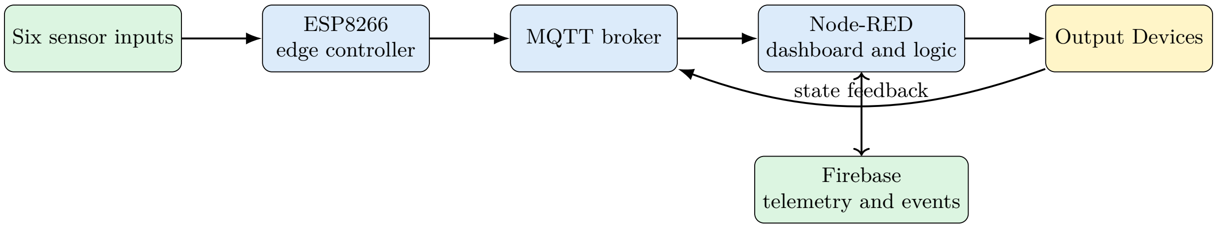

Table 1: Unified system inputs and outputs

Type	Component or parameter	Purpose
Input	Water pH	Detects acid or alkaline conditions.
Input	Dissolved oxygen	Detects oxygen shortage and controls aeration.
Input	Water temperature	Detects thermal stress and supports compensation.
Input	Water level	Detects overflow and low-level conditions.
Input	Rain	Detects rainfall and supports the heavy-rain workflow.
Input	Electrical conductivity	Measures raw EC and supports a calibrated salinity estimate.
Output	Aerator	Increases water circulation and dissolved oxygen.
Output	Pumping system	Removes excess water when overflow risk is detected. Introduces settling-pond water for salinity dilution.
Output	Buzzer and warning LED/beacon	Provides an immediate local warning.
Output	Feeder	Delivers a timed feed dose and stops during unsafe conditions.

4 | Detailed Description of Product Functions

4.1 | Proposed Operating Thresholds

The following values are used as configurable thresholds for the prototype, aligned with aquaculture guidelines [5] and international shrimp aquaculture water quality guidelines [6].

Table 2: Proposed operating thresholds for brackish water shrimp aquaculture

Parameter	**Unit**	**Normal range**	**Warning condition**
pH	–	7.5–8.5	pH < 7.5 (acidification) or > 8.8
Dissolved oxygen (DO)	mg/L	≥ 5.0	DO < 4.0 (hypoxia risk; < 3.5 critical)
Water temperature	°C	28–32	Temperature < 25°C or > 33°C
Salinity (EC K=1)	ppt ($\%_0$)	10–25	Salinity < 5 ppt or > 30 ppt (osmotic shock)
Water level	%	40–80	Water level $> 90\%$ (overflow) or $< 30\%$ (leak/evaporation)
Precipitation (Rain)	State	Dry / No Rain	Heavy rainfall detected (stratification & pH drop risk)

The thresholds can be changed from the Node-RED dashboard during testing.

4.2 | Environmental Monitoring

The Smart Shrimp Pond continuously monitors the main environmental parameters that affect shrimp health and pond stability. The proposed system uses six sensor inputs: **pH, dissolved oxygen (DO), water temperature, water level, rainfall, and electrical conductivity (EC)**, which is used to estimate salinity.

Together, these measurements provide an overview of the current pond condition. Rather than relying only on periodic manual measurements, the system provides continuous monitoring so that sudden environmental changes can be detected and reported to the farmer.

The measured values are displayed on the Node-RED dashboard and stored in Firebase for historical observation. The measurements also provide the environmental information required by the automatic response workflows described in Section 5.

4.3 | Environmental Status Classification

The system classifies the monitored pond conditions according to the operating thresholds defined in Section 4.1. Three status levels are used to provide a simple indication of pond condition.

Table 3: Environmental status levels

State	**Meaning**	**Color**
Normal	The monitored parameter is within the expected operating range.	Green
Warning	The monitored parameter has moved outside the preferred range and may require attention.	Yellow
Critical	The monitored condition presents a significant risk and requires an immediate response.	Red

The classification provides the farmer with a quick overview of pond conditions without requiring every sensor value to be interpreted individually. When abnormal conditions are detected, the corresponding status and warning are displayed on the dashboard and may also trigger an automatic response.

4.4 | Automatic and Manual Operating Modes

The proposed system supports both automatic and manual operation. These modes allow routine emergency responses to be automated while preserving direct control by the farmer when necessary.

4.4.1 | Automatic Mode

In automatic mode, the system observes the environmental measurements and responds to predefined hazardous conditions according to the configured thresholds.

The proposed automatic responses focus on three major pond-management situations:

1. Low dissolved oxygen and hypoxia
2. Heavy rainfall, acidification, and excessive water level
3. High temperature and salinity.

Depending on the detected condition, the system can operate the aerator, drainage pump, dilution intake pump, feeder, and local warning devices. Remote alerts are also provided so that the farmer remains informed when an automatic response occurs.

The purpose of automatic mode is not to replace the farmer's decision-making, particularly for treatments involving chemicals or other interventions that depend on actual pond conditions. Instead, it provides immediate responses to conditions where delayed action could increase the risk to the shrimp.

4.4.2 | Manual Mode

Manual mode allows the farmer to directly operate the controllable devices from the Node-RED dashboard.

The proposed manual controls include:

Aerator | **Drainage Pump** | **Dilution Intake Pump** | **Feeder**

The dashboard also displays the current state of the controlled devices, allowing the farmer to verify whether an actuator is operating.

Manual control is particularly useful during maintenance, testing, or situations where the farmer determines that a different response is more appropriate than the automatic workflow.

5 | Main Operating Scenarios

The proposed system combines individual sensor measurements into several practical pond-management scenarios. These scenarios represent the main environmental risks identified for the project and demonstrate how monitoring, warning, and control functions operate together.

5.1 | Scenario 1: Low Dissolved Oxygen

Dissolved oxygen is one of the primary parameters monitored by the system. A prolonged decrease in DO can expose shrimp to hypoxic conditions and therefore requires a rapid response.

The emergency condition is identified when:

$$\mathrm{DO} < 4.0\,\mathrm{mg/L} \quad \text{for at least 30 seconds}$$

When this condition is confirmed, the proposed system:

1. Classifies the pond condition as critical;
2. Activates the aerator to increase oxygenation and water circulation;
3. Suspends automatic feeding while the pond remains under hypoxic conditions;
4. Activates the local warning beacon and buzzer;
5. Sends a remote warning to the farmer; and
6. Records the event for later review.

The automatic response continues while the dissolved oxygen level remains unsafe. Normal operation can resume after the DO level has stabilized above:

$$\text{DO} > 5.5\,\text{mg/L}$$

This separation between the activation and recovery levels helps avoid repeated switching when the measured DO value fluctuates close to the emergency threshold.

5.2 | Scenario 2: Heavy Rainfall, Acidification, and High Water Level

Heavy rainfall can rapidly change the surface condition of a shrimp pond. Rainwater may increase the water level while also contributing to surface-water dilution, stratification, and changes in pH.

The rain sensor therefore works together with the water-level and pH measurements to identify potentially hazardous rainfall conditions.

An overflow condition is identified when:

$$\text{Water Level} > 90\% \quad \text{for at least 10 seconds}$$

When heavy rainfall or an excessive water level is detected, the proposed system can:

1. Warn the farmer of the rainfall and possible overflow risk;
2. Activate the drainage pump when the water level becomes excessively high;
3. Operate the aerator to improve surface mixing and reduce stratification;
4. Activate the local warning devices for critical conditions;
5. Monitor the pH for signs of acidification; and
6. Record the warning and device activity.

The drainage response is intended to reduce excessive pond water and overflow risk. The pump is stopped after the water level returns to a safer operating level.

If the rainfall event is accompanied by abnormal pH, the system alerts the farmer so that an appropriate corrective treatment can be considered. Chemical treatment, including the application of $CaCO_3$, remains a farmer-controlled action because the required treatment depends on the actual pond conditions and the quantity of water being treated.

5.3 | Scenario 3: High Temperature and Salinity

High midday temperatures can increase evaporation and contribute to increasing salinity. The proposed system therefore considers water temperature together with the salinity estimate obtained from the EC sensor.

The expected water-temperature and salinity ranges are defined in Section 4.1. Conditions approaching or exceeding these ranges are displayed as warnings on the dashboard.

When high temperature is accompanied by elevated salinity, the proposed system can:

1. Warn the farmer of possible thermal and osmotic stress;
2. Activate the dilution intake pump to introduce water from the settling pond;
3. Reduce or suspend automatic feeding while environmental conditions remain unsuitable;
4. Display the affected temperature and salinity measurements prominently on the dashboard; and
5. Record the event and corresponding device activity.

This workflow uses both temperature and salinity information instead of responding to either measurement independently. The purpose is to identify conditions associated with heat-driven evaporation while avoiding unnecessary dilution when only a temporary temperature increase occurs.

6 | Detailed Description of Web Functions

The Node-RED web interface provides the main connection between the farmer and the Smart Shrimp Pond system. It combines environmental monitoring, warning information, historical observations, and device control in a single interface accessible through the Internet.

6.1 | Dashboard Overview

Category	**Displayed Information**
Environmental	pH, dissolved oxygen, water temperature, water level, rainfall status, and EC-based salinity.
System	Overall pond status, connection status, operating mode, actuator states, and recent warnings or emergency events.

The Normal, Warning, and Critical classifications defined in Section 4.3 are used to make abnormal conditions visually distinguishable from normal pond operation.

6.2 | Real-Time Monitoring

The dashboard displays the latest measurements received from the pond monitoring system. This allows the farmer to observe environmental changes remotely without relying exclusively on direct manual inspection.

Each monitored parameter is presented together with its measurement unit and current environmental status. Rainfall is represented by its current detection state, while EC measurements are used to provide the corresponding salinity information.

The real-time view allows the farmer to observe whether pond conditions are stable and whether an automatic response has been activated.

6.3 | Remote Device Control

The web interface provides manual controls for the principal output devices of the proposed system:

Aerator | **Drainage Pump** | **Dilution Intake Pump** | **Automatic Feeder**

These controls allow the farmer to intervene remotely when manual operation is required. The dashboard also displays the current device states so that the farmer can observe the operating condition of each controlled device.

Automatic operation remains available for the emergency workflows described in Section 5. The farmer can switch between automatic and manual operation according to the pond condition and operational requirements.

6.4 | Historical Data Visualization

The system stores environmental measurements so that recent and historical pond conditions can be reviewed through the dashboard.

Historical visualization includes the principal numerical measurements:

pH | **Dissolved Oxygen** | **Water Temperature** | **Water Level**

Electrical Conductivity | **Estimated Salinity**

Rainfall events can also be associated with the recorded environmental data, allowing changes in pH, water level, and other parameters to be compared with rainfall conditions.

The historical view is intended to help the farmer identify trends and review the environmental conditions surrounding previous warning events. Advanced analytical or prediction functions are outside the scope of the proposed system.

6.5 | Threshold Settings

The operating thresholds defined in Section 4.1 are configurable from the dashboard for testing and adjustment.

The configurable settings correspond to the monitored environmental parameters:

pH Limits | **DO Limits** | **Temperature Limits** | **Salinity Limits**
Water-Level Limits | **Rainfall Warning Conditions**

These settings determine when the system changes environmental status, issues warnings, or initiates the corresponding automatic response.

The values presented in Section 4.1 are the proposed initial settings for the prototype. They may be adjusted during system testing and calibration.

6.6 | Alert and Event History

The web interface provides a record of important environmental warnings and control events. This allows the farmer to review previous abnormal conditions instead of relying only on the current dashboard state.

The recorded information includes:

Event Type | **Environmental Measurement** | **Event Time**
Device Action | **Event Status**

The event history covers the major emergency situations described in Section 5, including hypoxia, rainfall and overflow conditions, abnormal pH, and temperature–salinity stress.

The proposed history function is intended for basic monitoring and review. Advanced functions such as predictive analysis, automatic diagnosis, or complex event management are outside the scope of the project.

7 | Product Function Summary

Table 4: Summary of Smart Shrimp Pond functions mapped to system requirements

ID	Function	Description	Type
F01	pH measurement	Reads analog voltage from DFRobot pH Pro V2 and publishes pond pH to Node-RED via MQTT.	Input
F02	Dissolved oxygen (DO) monitoring	Reads DFRobot Analog DO Sensor Kit to track oxygen concentration (mg/L) for hypoxia detection.	Input
F03	Water temperature tracking	Converts digital DS18B20 signals into thermal data (°C) for midday heat and stress monitoring.	Input
F04	Water level measurement	Uses JSN-SR04T V3 ultrasonic sensor to measure pond surface height and detect overflow thresholds.	Input

ID	Function	Description	Type
F05	Rain detection	Detects precipitation events via analog rain sensor to predict surface stratification and acidification.	Input
F06	Salinity / EC estimation	Measures electrical conductivity (DFRobot EC V2) to estimate salinity (ppt) and evaporation spikes.	Input
F07	Surface aeration control	Actuates paddlewheel aerators via relay from Node-RED (manual/auto) and returns state feedback to MQTT.	Output
F08	Surface drainage pump control	Operates R385 12V DC surface pumps to discharge excess acidic rainwater and confirms relay state.	Output
F09	Dilution intake pump control	Triggers intake pumps to draw settling pond water when high temperature and salinity are detected.	Output
F10	Automatic feeder scheduling	Operates shrimp feeder and automatically halts feeding during hypoxia (DO < 4.0 mg/L).	Output
F11	Local visual & audio alarm	Activates LTE-1101J red beacon and MKE-M03 buzzer during emergency water quality breaches.	Output
F12	Real-time monitoring dashboard	Displays live sensor telemetry (pH, DO, Temp, Level, Rain, EC) on the Node-RED web interface via Internet.	Web
F13	Remote actuator control interface	Provides web-based toggle switches and buttons to manually override paddlewheels, pumps, and feeders remotely.	Web
F14	Historical data visualization	Renders interactive time-series charts and graphs on the website using historical sensor data queried from Firebase.	Web
F15	Sensor telemetry logging	Node-RED logs continuous time-series data (pH, DO, EC, Temp, Level, Rain) to Firebase for historical analysis.	Storage
F16	Actuator event history	Records all manual commands, automated relay triggers, and ESP8266 state feedback logs to Firebase cloud.	Storage
F17	Hypoxia emergency alert	Dispatches immediate mobile notification/email when DO < 4.0 mg/L and triggers aeration workflow.	Alert
F18	Acidification & flood alert	Sends mobile app/email alerts to prepare $CaCO_3$ liming when rain and high water levels are detected.	Alert
F19	Salinity & thermal shock alert	Notifies farmers of midday evaporation spikes ($> 33°C$, > 30 ppt) and prompts feed reduction.	Alert

8 | List of Components

Table 5: Bill of Materials for the Shrimp Pond IoT System

Category	Component	Qty.	Unit Price (VND)	Total (VND)	Source
Controller	ESP8266 NodeMCU Lua V3 CH340	1	72,000	72,000	HShop
Sensor	Rain Sensor Module	1	11,000	11,000	HShop
Sensor	DFRobot Gravity Analog pH Sensor Pro V2	1	1,674,000	1,674,000	HShop
Sensor	MKE-S15 DS18B20 Waterproof Temperature Sensor	1	75,000	75,000	HShop
Sensor	JSN-SR04T V3 Ultrasonic Sensor	1	155,000	155,000	HShop
Sensor	DFRobot Gravity Analog Dissolved Oxygen Sensor Kit	1	4,644,000	4,644,000	HShop
Sensor	DFRobot Gravity Analog EC Sensor V2 (K=1)	1	1,836,000	1,836,000	HShop
Output	8-Channel 5V Opto-Isolated Relay Module	1	85,000	85,000	Vietnic
Output	R385 12V DC Water Pump	2	25,000	50,000	HShop
Output	2NGUYEN Automatic Shrimp Feeder	1	3,375,000	3,375,000	Khai Phat
Output	MKE-M03 Buzzer Module	1	25,000	25,000	HShop
Output	LTE-1101J 12V Red Warning Beacon	1	190,000	190,000	Khánh An CCTV
			Required Components Total	**12,192,000**	
Optional	Paddle-Wheel Aerator (Prototype/CAD Estimate)	1	800,000	800,000	Commercial Reference
			Total Including Optional Aerator	**12,992,000**	

9 | Product Sketch

9.1 | Shape, external outline

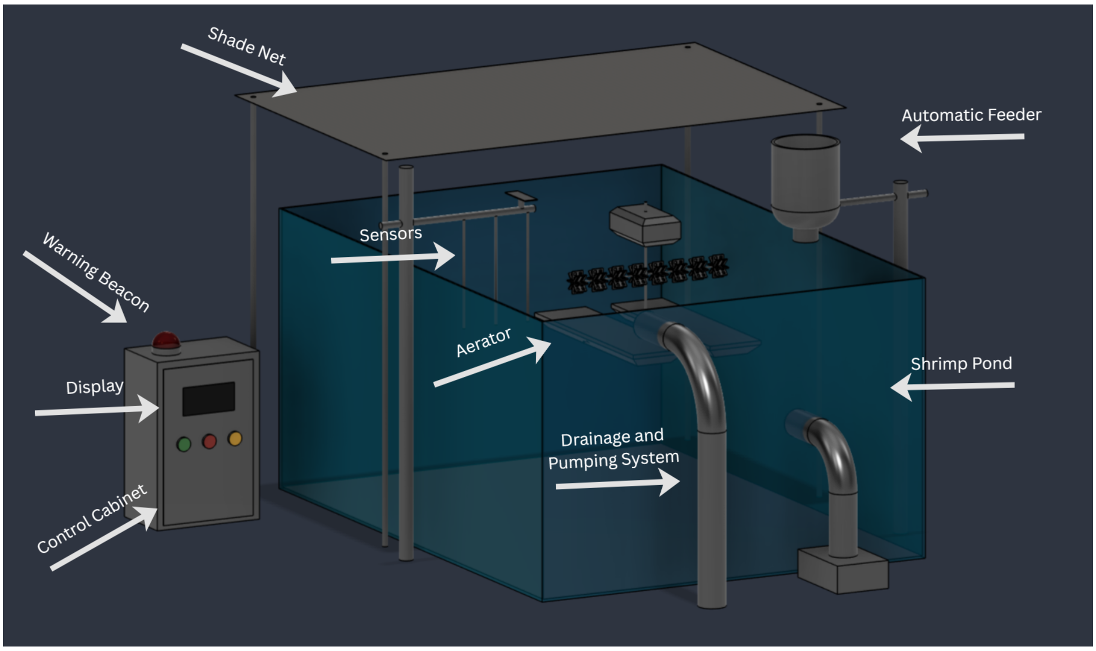

Figure 1: 3D model of a smart shrimp farming pond monitoring and control system

9.2 | Internal outline

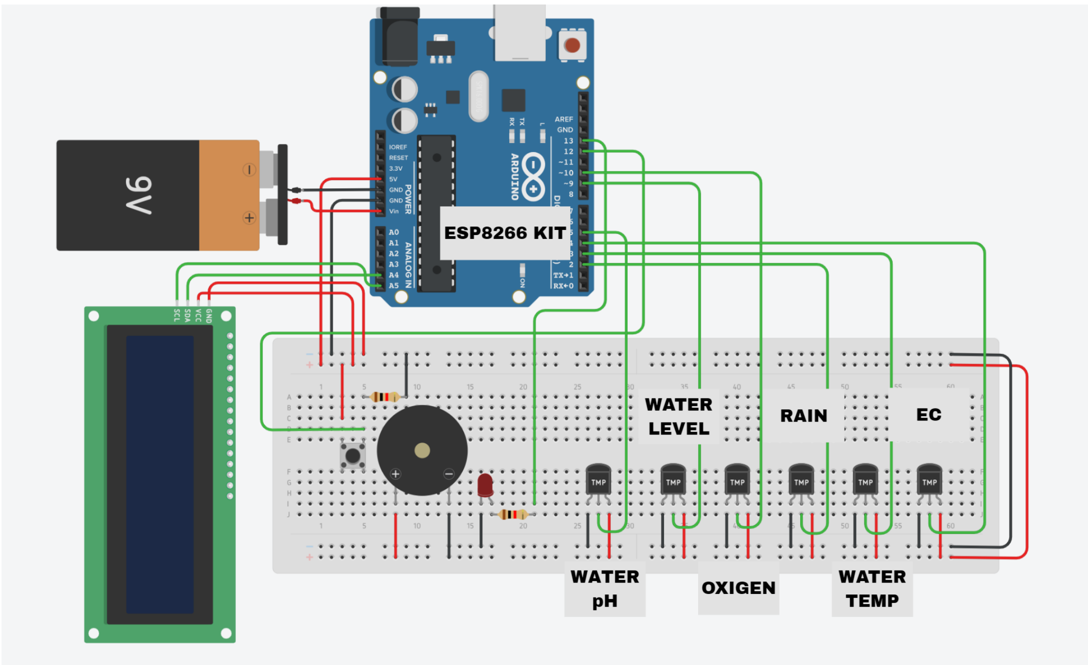

Figure 2: Electrical wiring schematic of the Smart Shrimp Pond

10 | Project Plan

10.1 | Detailed Week-by-Week Plan

Base on the feature seperation above, the project plan devide into weeks step by step mileston to finish the product.

No.	Start Date	End Date	Task Description	Assigned To
1	20/07/2026	21/07/2026	Project planning and task allocation	Huynh Tham
2	20/07/2026	23/07/2026	Define product description, core workflow of system	Huynh Tham
3	22/07/2026	25/07/2026	Define system requirements, research on practical demands	Hoai Phuc
4	24/07/2026	26/07/2026	List required hardware components, sensors with pricing and suppliers	Cao Thong
5	25/07/2026	26/07/2026	Design hardware architecture, draw product design on AutDDesk Fusion	Cao Thong
6	27/07/2026	29/07/2026	Give detail functions for product	Hoai Phuc
7	27/07/2026	30/07/2026	Select cloud platform and deploy Node-RED / MQTT server	Huynh Tham
8	29/07/2026	02/08/2026	Design and program web dashboard UI/UX	Hoai Phuc
9	30/07/2026	02/08/2026	Program server communication protocols and APIs	Huynh Tham
10	03/08/2026	06/08/2026	Program sensor data acquisition firmware (ESP8266)	Cao Thong
11	03/08/2026	06/08/2026	Program actuator control logic (relays, pumps, aerators)	Cao Thong
12	06/08/2026	08/08/2026	Implement emergency alert notification via Email	Huynh Tham
13	07/08/2026	09/08/2026	Set up cloud data logging and historical chart synchronization	Hoai Phuc
14	10/08/2026	12/08/2026	Write documentation and presentation outline	Cao Thong
15	12/08/2026	15/08/2026	End-to-end system testing, calibration, and bug fixing	Hoai Phuc
16	15/08/2026	17/08/2026	Finalize documentation, slides, and project report	Huynh Tham

Table 6: IoT Project Implementation Schedule (20/07/2026 – 17/08/2026)

To ensure balanced collaboration, the 16 project tasks are divided among the three team members based on their technical specialization, with each task assigned to exactly one primary owner:

- **Ngo Huynh Tham (34%)**
- **Hoai Phuc (33%)**
- **Cao Thong (33% – Tasks 4, 5, 10, 11, 14):** Responsible for hardware BOM procurement, AutDDesk Fusion CAD enclosure design, ESP8266 sensor/actuator C++ firmware programming, and drafting technical hardware documentation.

10.2 | Detailed Task Execution Plan

The following section outlines the execution methodology, scope, and required deliverable for each task in the project schedule.

Week 1: Planning, Requirements, and Hardware Architecture (20/07 – 26/07/2026)

Task 1: Project planning and task allocation (Huynh Tham)
Scope: Establish communication channels, set up version control repositories (GitHub/Google Drive), and define project milestones.
Deliverable: Finalized Gantt chart and signed team charter.

Task 2: Define product description and core system workflow (Huynh Tham)
Scope: Formulate the 3 core closed-loop automation workflows (Rain/pH, DO/Hypoxia, Heat/Salinity) and map the IoT data pipeline.
Deliverable: System architecture block diagram and workflow description chapter.

Task 3: Define system requirements and research practical demands (Hoai Phuc)
Scope: Survey brackish water aquaculture guidelines (QCVN 02-19:2014) to establish numerical operating thresholds for pH, DO, Temp, and Salinity.
Deliverable: Functional and non-functional requirements specification document.

Task 4: List required hardware components and suppliers (Cao Thong)
Scope: Source sensors (DFRobot pH/DO/EC, DS18B20, JSN-SR04T), controller (ESP8266), actuators, and relays; verify voltage/current compatibility.
Deliverable: Finalized Bill of Materials (BOM) with unit prices and Vietnamese vendor links.

Task 5: Design hardware architecture and CAD prototype (Cao Thong)
Scope: Draw electrical wiring schematics and design the waterproof control box enclosure using Autodesk Fusion.
Deliverable: Complete wiring schematic diagram and 3D CAD enclosure models.

Week 2: Procurement, UI/UX, and Cloud Platform Setup (27/07 – 02/08/2026)

Task 6: Detail product functions mapped to requirements (Hoai Phuc)
Scope: Catalog all 19 system functions across Input, Output, Web, Storage, and Alert categories.
Deliverable: Fully documented Product Function Summary table.

Task 7: Select cloud platform and deploy Node-RED / MQTT broker (Huynh Tham)
Scope: Provision cloud server hosting, install Eclipse Mosquitto MQTT broker, and configure Node-RED container environment.
Deliverable: Operational cloud broker accepting authentication credentials.

Task 8: Design and program web dashboard UI/UX (Hoai Phuc)
Scope: Build the Node-RED web interface containing live sensor gauges, manual relay control toggle switches, and status alerts.
Deliverable: Interactive web dashboard accessible via Internet browser.

Task 9: Program server communication protocols and APIs (Huynh Tham)
Scope: Define JSON telemetry payload schemas for MQTT (‘aqua/sensors‘, ‘aqua/control‘, ‘aqua/state‘) and configure Node-RED parsing flows.
Deliverable: Tested MQTT JSON payload specifications and backend router flows.

Week 3: Firmware Programming and Cloud Sync (03/08 – 09/08/2026)

Task 10: Program sensor data acquisition firmware (Cao Thong)
Scope: Write C++ firmware for ESP8266 in Arduino IDE to read analog voltages, apply calibration curves (pH/DO/EC), and sample DS18B20/ultrasonic data.
Deliverable: Compiled sensor acquisition firmware module.

Task 11: Program actuator control logic and state feedback (Cao Thong)
Scope: Implement GPIO relay control routines for aerators, pumps, and automatic feeders; program MQTT state feedback reporting after actuation.
Deliverable: Actuator firmware module with edge-level emergency override logic.

Task 12: Implement emergency alert notification via Email (Huynh Tham)
Scope: Configure Node-RED email/notification nodes (SMTP/Push) to trigger alerts whenever DO $<$ 4.0 mg/L or pH $<$ 7.5.
Deliverable: Automated alert engine verified via test email dispatches.

Task 13: Set up cloud data logging and historical chart sync (Hoai Phuc)
Scope: Connect Node-RED to Google Firebase Realtime Database to store time-series sensor logs and render historical charts on the dashboard.
Deliverable: Live database logging pipeline and web dashboard history view.

Week 4: Assembly, Testing, and Documentation (10/08 – 17/08/2026)

Task 14: Write documentation and presentation outline (Cao Thong)
Scope: Draft technical hardware documentation, wiring guides, and prepare the slide deck outline for project defense.
Deliverable: Drafted technical chapters and slide presentation skeleton.

Task 15: End-to-end system testing, calibration, and bug fixing (Hoai Phuc)
Scope: Conduct physical water bucket testing; calibrate DFRobot pH/DO sensors with buffer solutions; simulate rain and hypoxia to verify relays.
Deliverable: System test report, calibration error logs, and stable V1.0 release.

Task 16: Finalize documentation, slides, and project report (Huynh Tham)
Scope: Assemble all LaTeX report sections, check IEEE formatting/bibtex citations, review final defense slides, and verify GitHub code release.
Deliverable: Final PDF project report, defense presentation slides, and submitted code repository.

11 | Evaluation Against Course Requirements

No.	Course Requirement	Product Implementation & Evidence	Status
1	**ESP8266 Microcontroller**	Uses ESP8266 NodeMCU Lua V3 CH340 with Wi-Fi/Internet connectivity.	**Passed**
2	**$\geq$ 1 INPUT and 1 OUTPUT**	**Exceeded:** 6 INPUT sensors (pH, DO, Temp, Level, Rain, EC) and 5 OUTPUT devices (Aerator, Pumps, Feeder, Alarms).	**Passed**
3	**Web Internet Monitoring**	Real-time telemetry from all 6 sensors displayed on an interactive Node-RED web dashboard over the Internet.	**Passed**
4	**Web Internet Control**	Web interface provides remote toggle switches for aerators, drainage/dilution pumps, and automatic feeders.	**Passed**
5	**Cloud Data Storage**	Continuous time-series sensor data and actuator event history logged to Google Firebase Realtime Database.	**Passed**
6	**Emergency Notifications**	Automated mobile/email alerts dispatched during hypoxia (DO $<$ 4.0 mg/L), acidification, and overflow risks.	**Passed**
7	**Node-RED / Custom Code Only**	Management dashboard built exclusively with Node-RED; embedded edge controller coded manually in C++.	**Passed**
8	**Member Programming Participation**	All 3 members program: **Tham** (Node-RED backend/email alerts), **Phuc** (Web UI/Firebase sync), **Thong** (C++ ESP8266 firmware).	**Passed**

Table 7: Evaluation checklist of mandatory course project requirements

References

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[3] S. Patkaew, S. DirekbusarakoKm, I. Hirono, S. Wuthisuthimethavee, S. Powtongsook, and C. Pooljun, "Effect of supersaturated dissolved oxygen on growth-, survival-, and immune-related gene expression of pacific white shrimp (litopenaeus vannamei)," *Veterinary World*, pp. 50–58, 2024. DOI: 10.14202/vetworld.2024.50-58

[4] N. Thadtapong, M. B. S. Salinas, V. Charoensawan, V. Saksmerprome, and S. Chaturongakul, "Genome characterization and comparison of early mortality syndrome causing vibrio parahaemolyticus pirabvp mutant from thailand with v. parahaemolyticus pirabvp+ ahpnd isolates," *Frontiers in Marine Science*, vol. 7, 2020. DOI: 10.3389/fmars.2020.00290

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[6] Food and Agriculture Organization of the United Nations, *Farming the Marine Shrimp Litopenaeus vannamei.* FAO, 2022.